

LAB EXPERIMENT RECORD**03-10-2026---Set 8: Employee Performance Analysis**

Department	AIML	
Course Code / Course Name	DSA0601 / Data Handling and Visualization	
Experiment	Data Visualization using R and Tableau	
Name	DUDEKULA MOHAMMAD ILYAS	
Register Number	192325017	
Date	03 OCTOBER 2026	
Day	Saturday	

Aim

To analyze employee performance data using a line chart, bar chart and table, and to build an interactive Tableau dashboard for exploring employee performance.

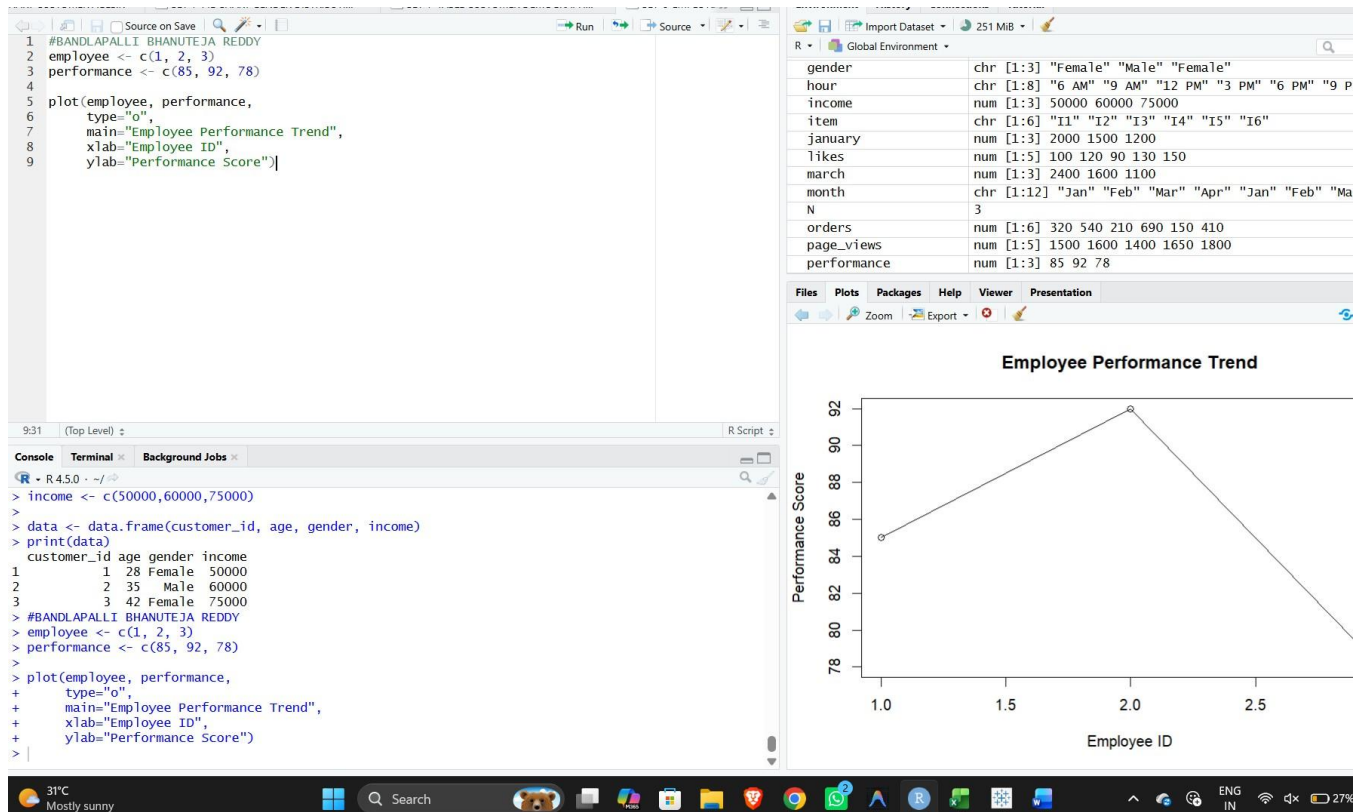
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Question 1: Line Chart of Performance Trend

Question: Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Aim: To visualize the performance trend of employees using a line chart with labelled axes and a title.

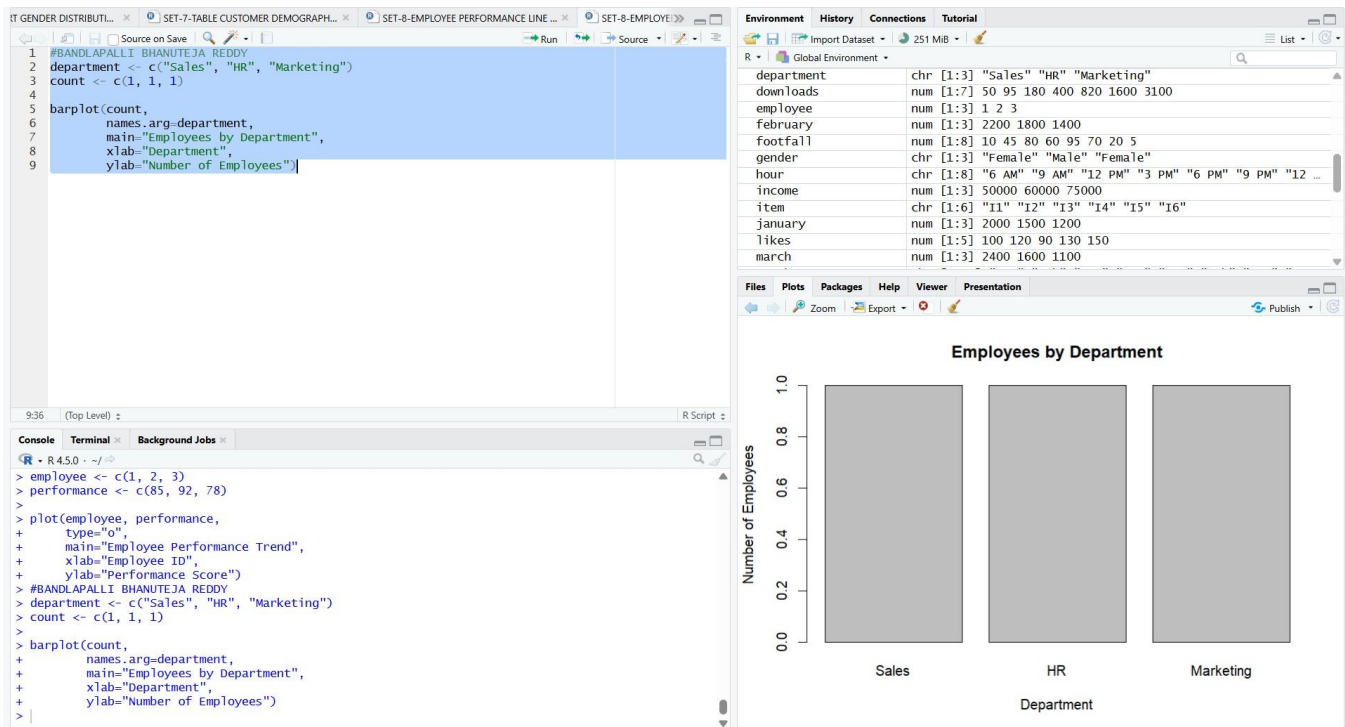


Result: Performance rises from 85 for Employee 1 to a peak of 92 for Employee 2, then falls to the lowest value of 78 for Employee 3.

Question 2: Bar Chart of Employees by Department

Question: Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Aim: To show the distribution of employees across departments using a labelled bar chart.



Result: Sales, HR and Marketing each have exactly 1 employee, so the three bars are equal in height.

Question 3: Table of Employee Performance

Question: Build a table to display the performance data for each employee. Label the table elements.

Aim: To present the department, years of service and performance score of each employee in a labelled table.



```
1 #BANDLAPALLI BHANUTEJA REDDY
2 employee_id <- c(1, 2, 3)
3 department <- c("Sales", "HR", "Marketing")
4 years <- c(5, 3, 7)
5 performance <- c(85, 92, 78)
6
7 data <- data.frame(employee_id, department, years, performance)
8
9 print(data)|
```

9:12 (Top Level) ⬆

Console **Terminal** **Background Jobs**

R 4.5.0 · ~/

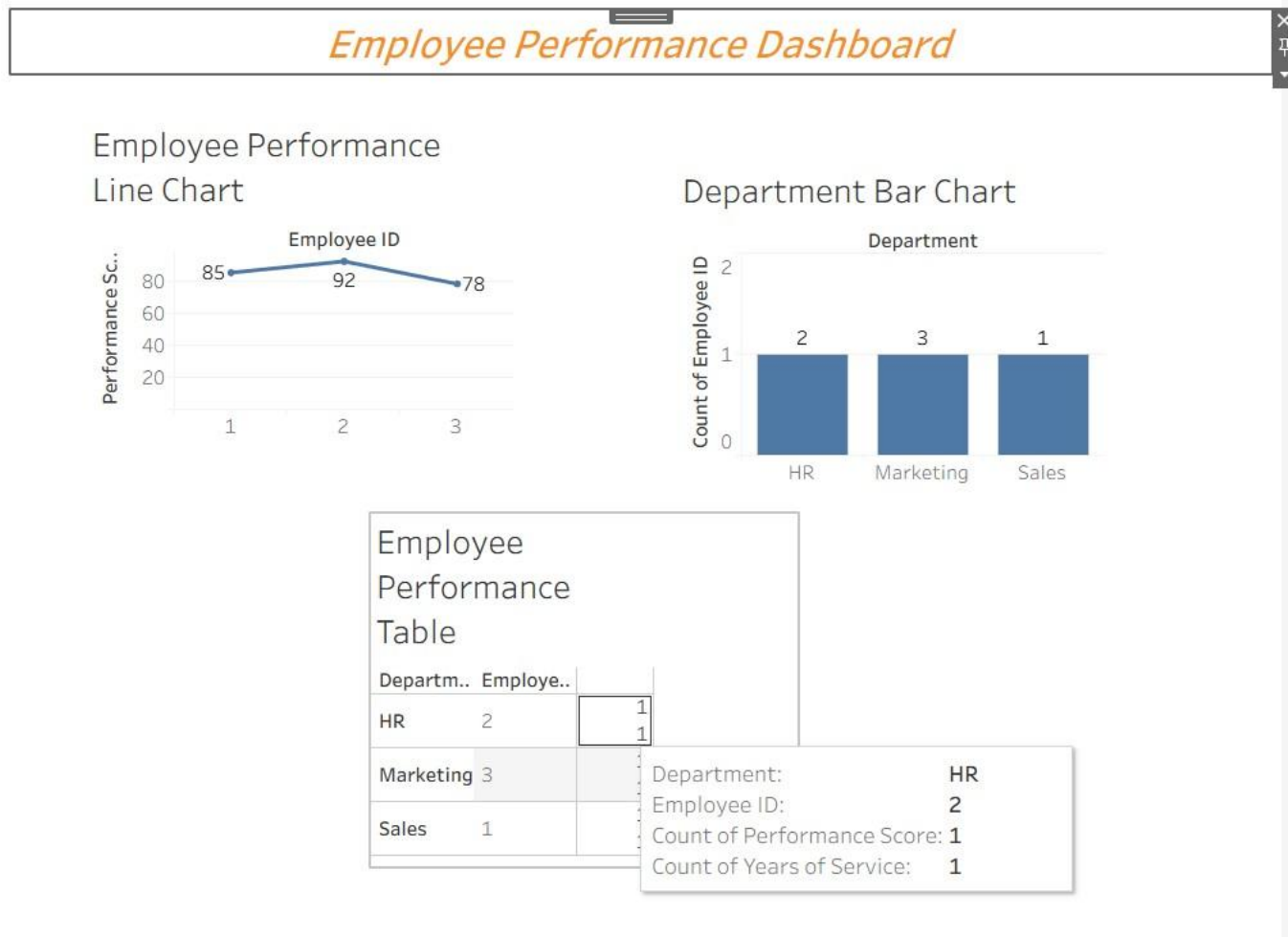
```
+      names.arg=department,
+      main="Employees by Department",
+      xlab="Department",
+      ylab="Number of Employees")
> #BANDLAPALLI BHANUTEJA REDDY
> employee_id <- c(1, 2, 3)
> department <- c("Sales", "HR", "Marketing")
> years <- c(5, 3, 7)
> performance <- c(85, 92, 78)
>
> data <- data.frame(employee_id, department, years, performance)
>
> print(data)
  employee_id department years performance
1           1     Sales     5           85
2           2        HR     3           92
3           3  Marketing     7           78
> |
```

Result: The table lists all three employees; the HR employee has the highest score (92), and the Marketing employee, with the most service (7 years), has the lowest (78).

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Aim: To build an interactive Tableau dashboard that combines the line chart, bar chart and table for exploring employee performance.



Result: The dashboard places all three views together, and a department filter updates them so users can explore performance interactively.

Overall Result

Observation: The average performance score is 85, and a longer service period does not guarantee a higher score in this small sample.